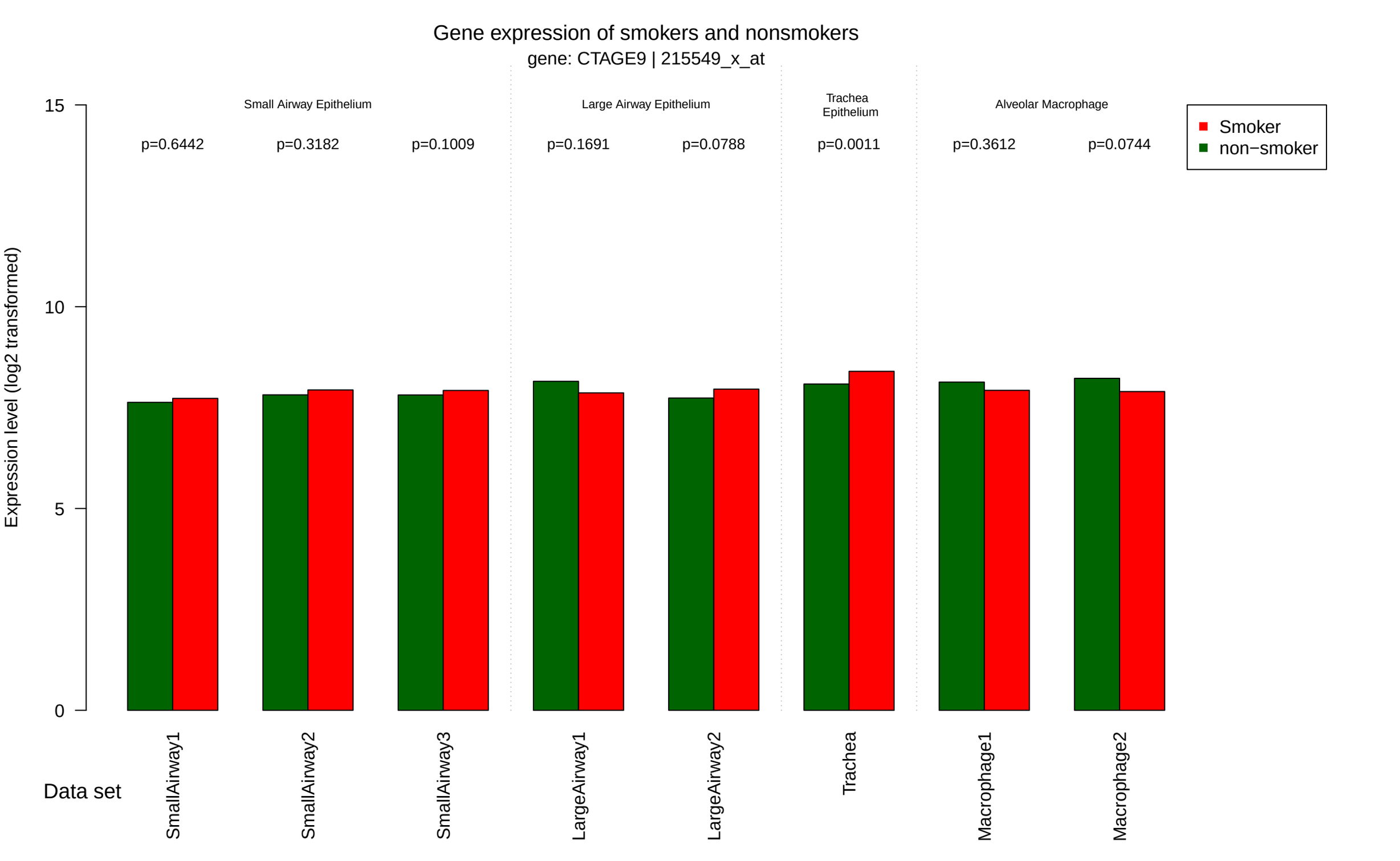
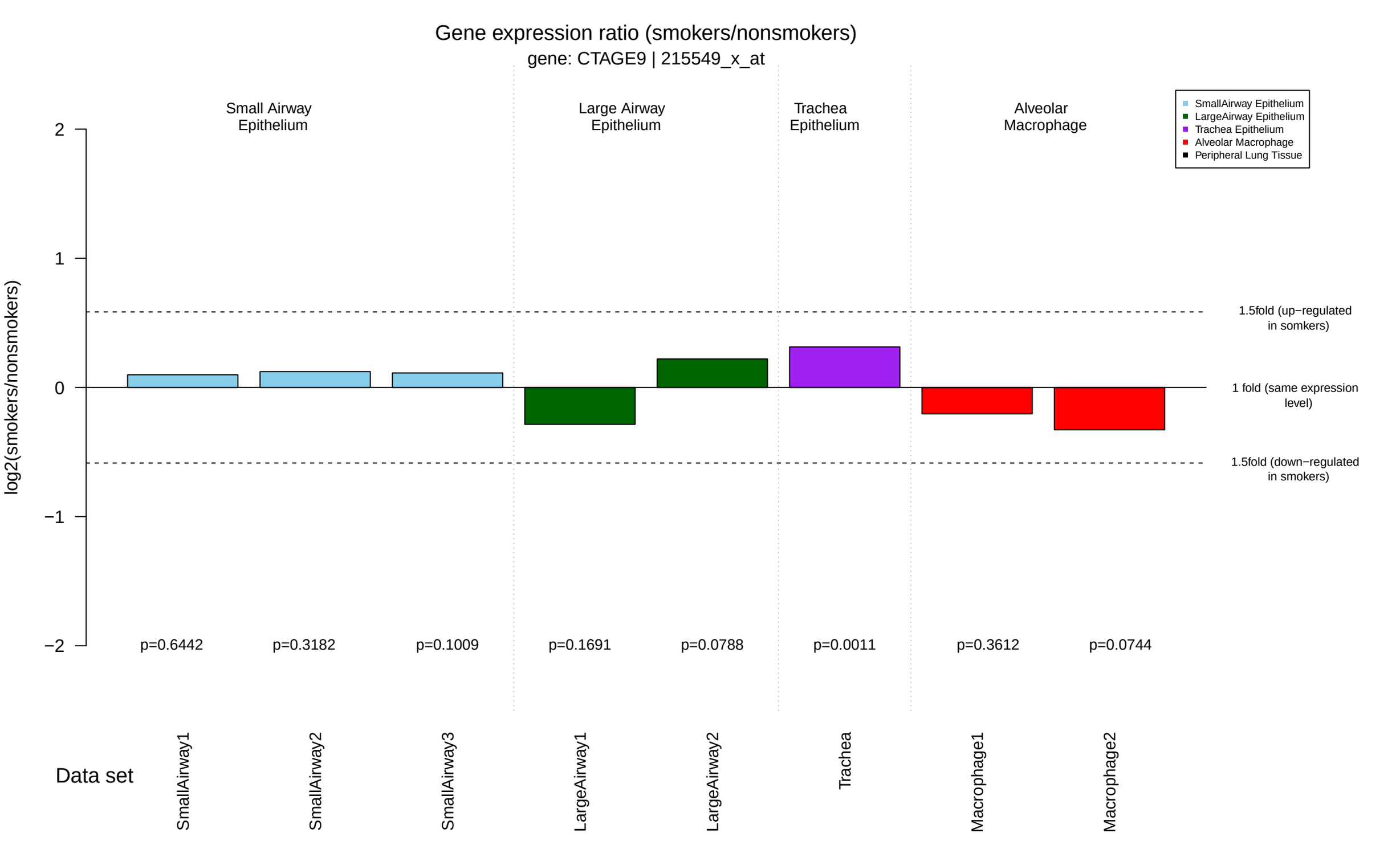
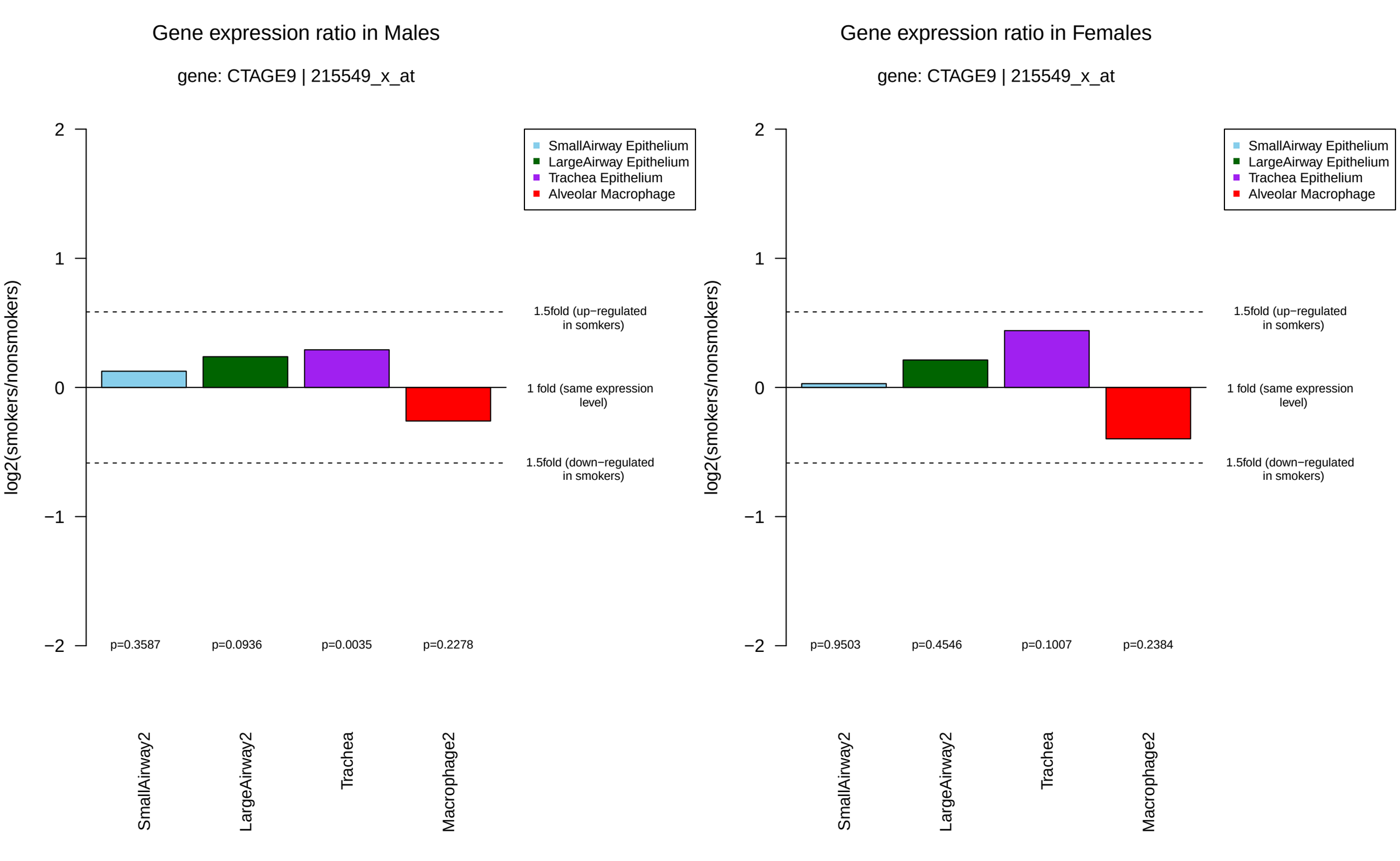
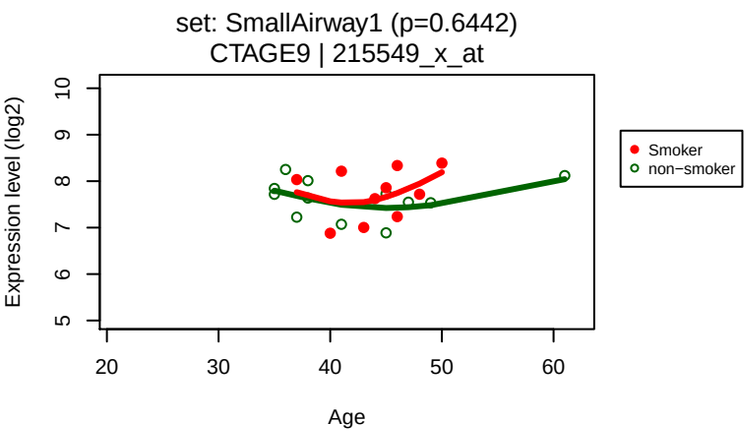
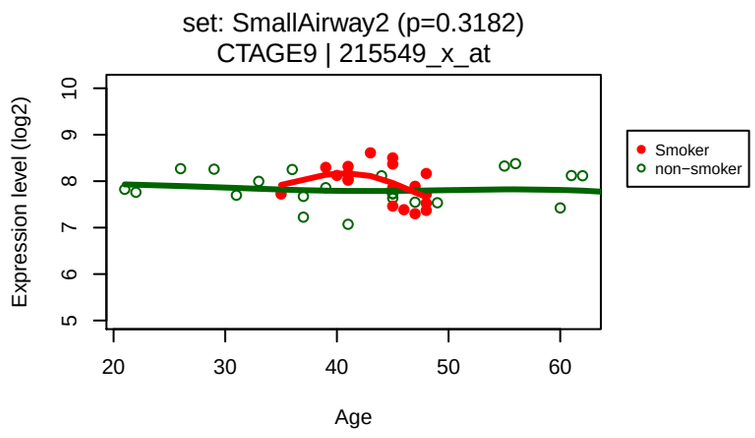
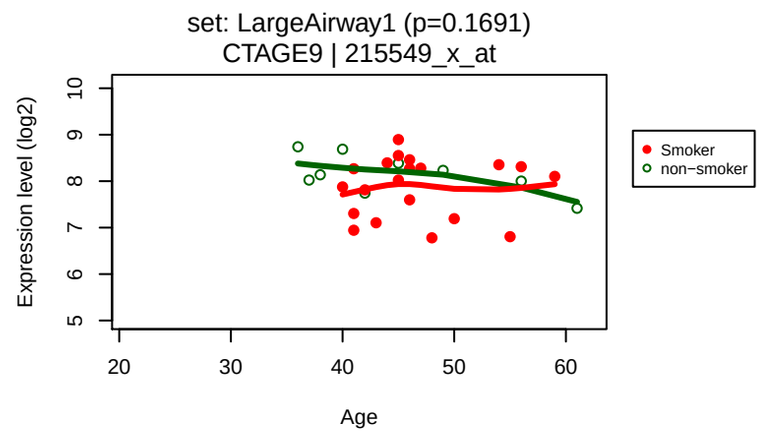

Gene expression of smokers and nonsmokers
gene: CTAGE9 | 215549_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6442
p=0.3182
p=0.1009
p=0.1691
p=0.0788
p=0.0011
p=0.3612
p=0.0744
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

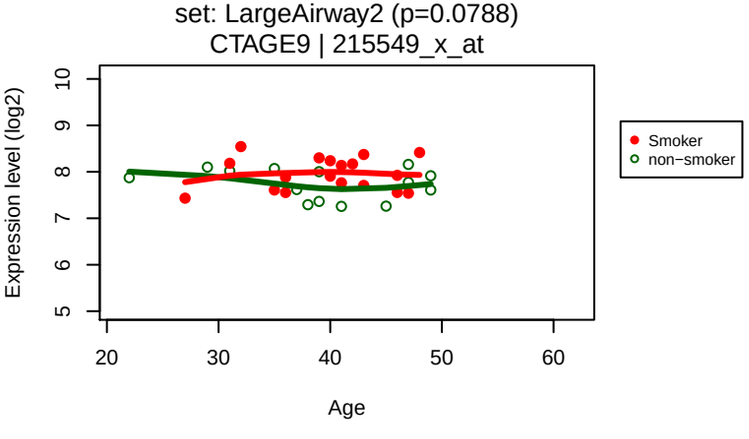

Gene expression ratio (smokers/nonsmokers)
gene: CTAGE9 | 215549_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6442
p=0.3182
p=0.1009
p=0.1691
p=0.0788
p=0.0011
p=0.3612
p=0.0744
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

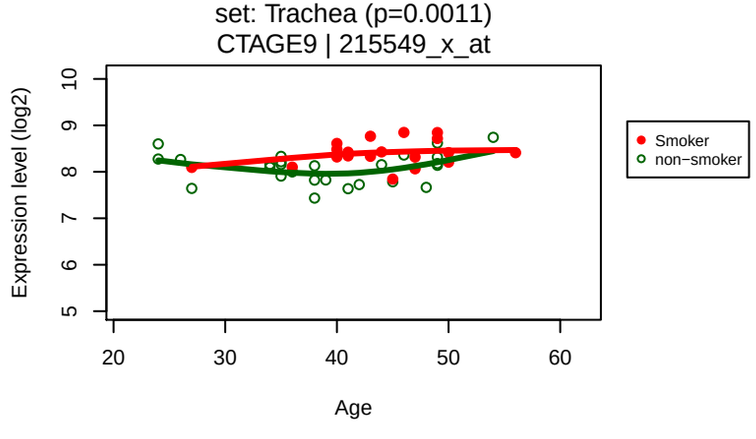

Gene expression ratio in Males
gene: CTAGE9 | 215549_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3587
p=0.0936
p=0.0035
p=0.2278
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CTAGE9 | 215549_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9503
p=0.4546
p=0.1007
p=0.2384
SmallAirway2
LargeAirway2
Trachea
Macrophage2

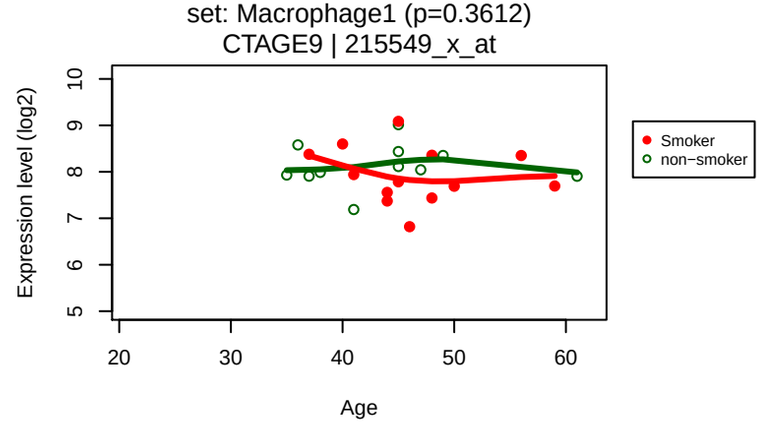

set: SmallAirway1 (p=0.6442)
CTAGE9 | 215549_x_at
Expression level (log2)
Age
Smoker
non−smoker

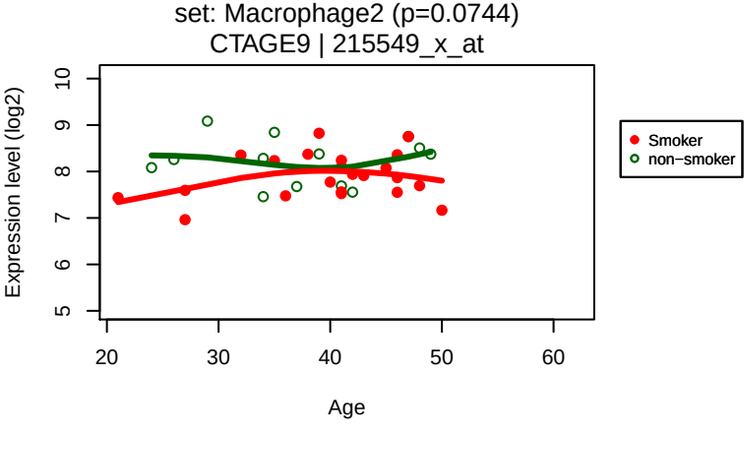

set: SmallAirway2 (p=0.3182)
CTAGE9 | 215549_x_at
Expression level (log2)
Age
Smoker
non−smoker

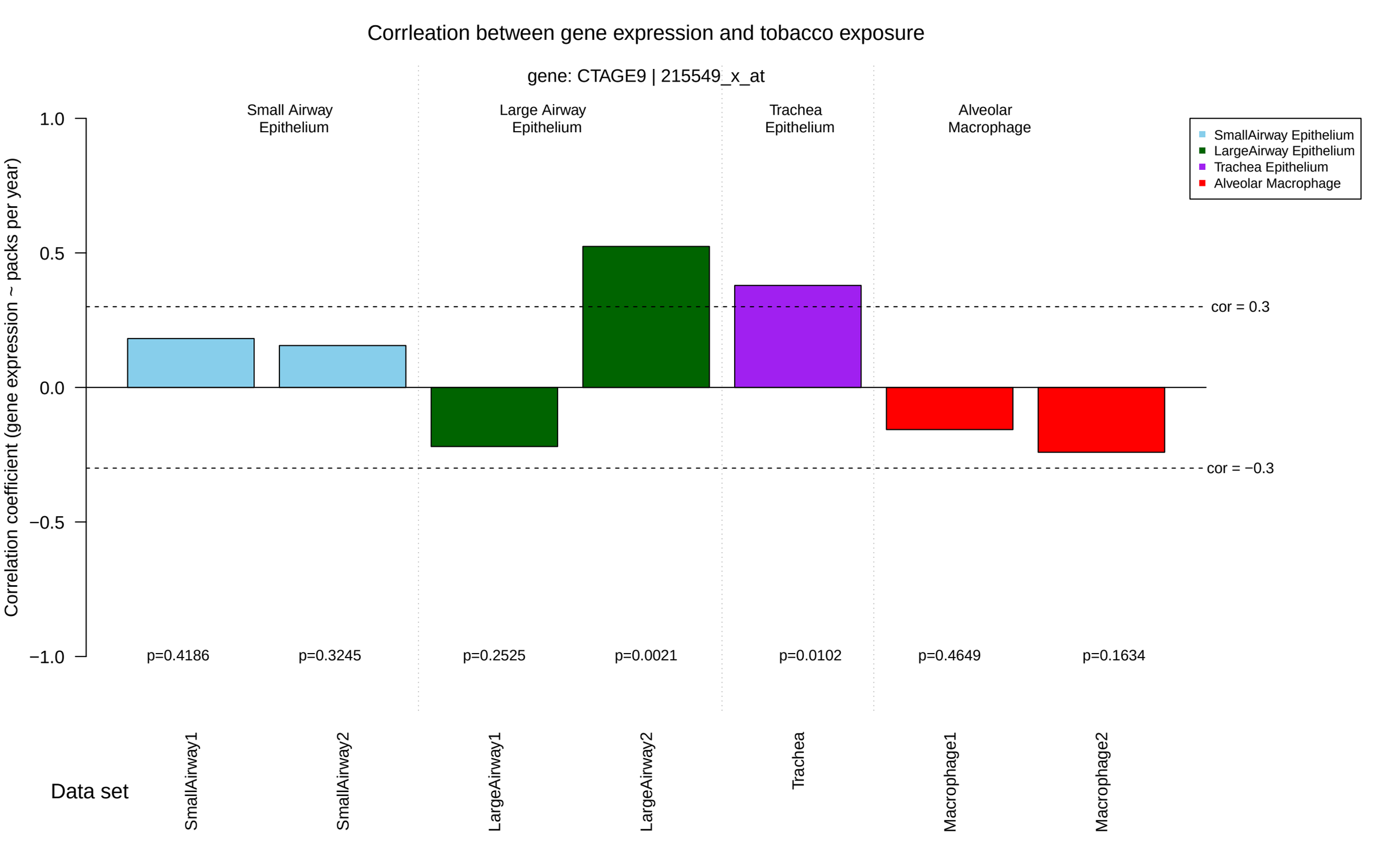

set: LargeAirway1 (p=0.1691)
CTAGE9 | 215549_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0788)
CTAGE9 | 215549_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0011)
CTAGE9 | 215549_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3612)
CTAGE9 | 215549_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0744)
CTAGE9 | 215549_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CTAGE9 | 215549_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4186
p=0.3245
p=0.2525
p=0.0021
p=0.0102
p=0.4649
p=0.1634
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2